

CHEM 101 Topic 12: Introduction to Organic Chemistry — Problem Set

Instructions: Complete the following problems using the analysis method practiced in your guided assignment. For hydrocarbon problems, use the general formulas provided in the lecture. For biomolecules, identify the primary function of the molecule.

Section 1: What Makes Chemistry "Organic"?

Problem 1: Determine if the following compounds are organic or inorganic: CH_4 (methane), CaCO_3 (calcium carbonate), $\text{C}_3\text{H}_7\text{OH}$ (propanol), and KI (potassium iodide).

Problem 2: Why is carbon uniquely suited to be the "backbone" of organic molecules compared to other elements?

Problem 3: Compare the typical bonding type and state of matter for an organic molecule versus an inorganic salt.

Section 2: Hydrocarbons

Problem 4: A hydrocarbon has the formula C_4H_{10} . Classify it as an alkane, alkene, or alkyne.

Problem 5: What is the name of a straight-chain alkane containing 3 carbon atoms?

Problem 6: Predict the molecular formula for an alkyne with 4 carbon atoms.

Section 3: Functional Groups

Problem 7: Identify the functional group present in the following molecule: $\text{CH}_3\text{CH}_2\text{CHO}$ (propanal).

Problem 8: Which functional group is characterized by the structure $-\text{COOH}$? Provide one example of a common substance that contains this group.

Problem 9: A drug molecule contains an $-\text{NH}_2$ group. What is the name of this functional group, and what is its general role in drug-receptor binding?

Section 4: Isomers and Biomolecules

Problem 10: Two molecules share the formula $\text{C}_3\text{H}_6\text{O}$ but have different boiling points and react differently. What is the term for these molecules?

Problem 11: Match the following biomolecules to their primary function: Carbohydrates, Lipids, Proteins, Nucleic Acids. (A) Genetic info, (B) Long-term energy, (C) Primary energy, (D) Enzymes/Structure.

Problem 12: Explain the relationship between the sequence of amino acids in a protein and its final biological function.